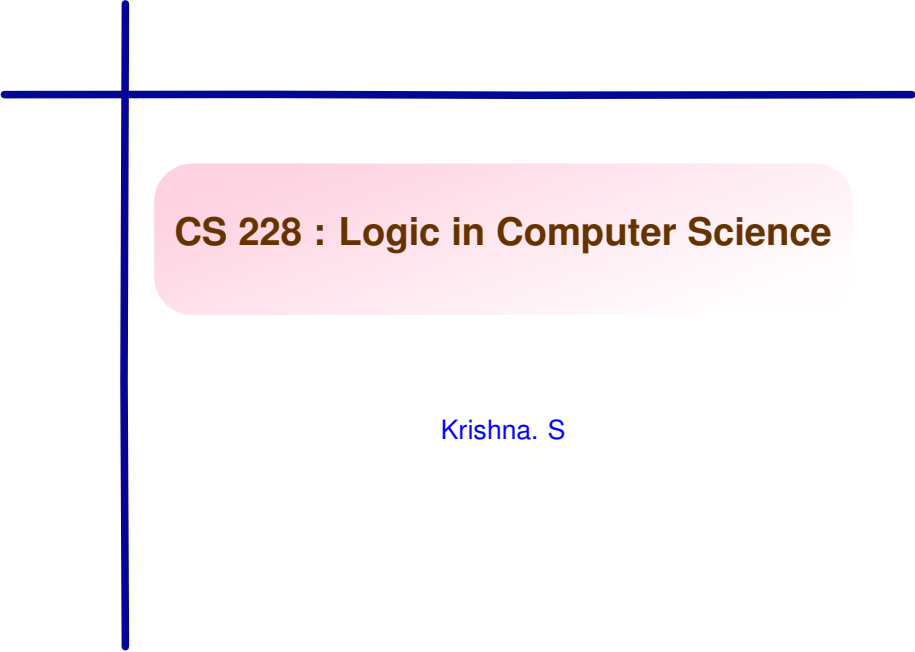


A decorative blue crosshair consisting of a vertical line and a horizontal line intersecting in the upper-left quadrant of the slide.

CS 228 : Logic in Computer Science

Krishna. S

Equivalence of LTL Formulae

Equivalence

φ and ψ are equivalent ($\varphi \equiv \psi$) iff $L(\varphi) = L(\psi)$.

Expansion Laws

- ▶ $\varphi \text{ U } \psi \equiv \psi \vee (\varphi \wedge \bigcirc(\varphi \text{ U } \psi))$
- ▶ $\diamond\varphi \equiv \varphi \vee \bigcirc\diamond\varphi$
- ▶ $\Box\varphi \equiv \varphi \wedge \bigcirc\Box\varphi$

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Distribution

$$\bigcirc(\varphi \vee \psi) \equiv \bigcirc\varphi \vee \bigcirc\psi,$$

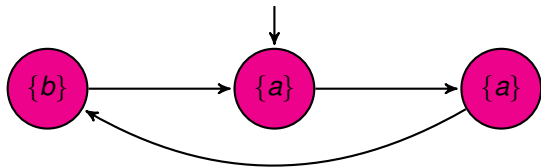
$$\bigcirc(\varphi \wedge \psi) \equiv \bigcirc\varphi \wedge \bigcirc\psi,$$

$$\bigcirc(\varphi \mathbf{U} \psi) \equiv (\bigcirc\varphi) \mathbf{U} (\bigcirc\psi),$$

$$\Diamond(\varphi \vee \psi) \equiv \Diamond\varphi \vee \Diamond\psi,$$

$$\Box(\varphi \wedge \psi) \equiv \Box\varphi \wedge \Box\psi$$

Equivalence of LTL Formulae



$$TS \models \Diamond a \wedge \Diamond b, TS \not\models \Diamond(a \wedge b)$$

$$TS \models \Box(a \vee b), TS \not\models \Box a \vee \Box b$$

Satisfiability, Model Checking of LTL

Two Questions

Given transition system TS , and an LTL formula φ . Does $TS \models \varphi$?

Given an LTL formula φ , is $L(\varphi) = \emptyset$?

ω -automata

An ω -automaton is a tuple $\mathcal{A} = (Q, \Sigma, \delta, q_0, Acc)$ where

- ▶ Q is a finite set of states
- ▶ Σ is a finite alphabet
- ▶ $\delta : Q \times \Sigma \rightarrow 2^Q$ is a state transition function (if non-deterministic, otherwise, $\delta : Q \times \Sigma \rightarrow Q$)
- ▶ $q_0 \in Q$ is an initial state and Acc is an acceptance condition

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Run

A run ρ of $\mathcal{A}$ on an ω -word $\alpha = a_1 a_2 \dots \in \Sigma^\omega$ is an infinite state sequence $\rho(0)\rho(1)\rho(2)\dots$ such that

- ▶ $\rho(0) = q_0$,
- ▶ $\rho(i) = \delta(\rho(i-1), a_i)$ if $\mathcal{A}$ is deterministic,
- ▶ $\rho(i) \in \delta(\rho(i-1), a_i)$ if $\mathcal{A}$ is non-deterministic,

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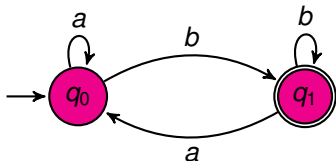
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Büchi Acceptance

For Büchi Acceptance, Acc is specified as a set of states, $G \subseteq Q$. The ω -word α is accepted if there is a run ρ of α such that $Inf(\rho) \cap G \neq \emptyset$.

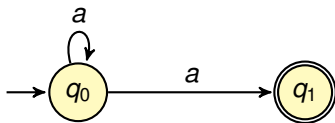
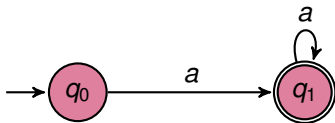
ω -Automata with Büchi Acceptance



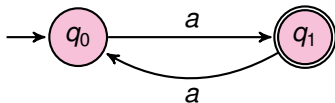
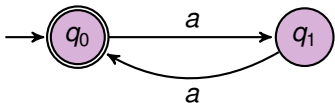
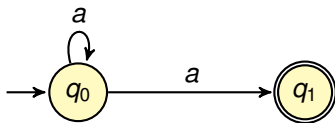
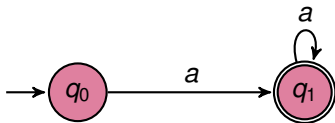
$$L(\mathcal{A}) = \{\alpha \in \Sigma^\omega \mid \alpha \text{ has a run } \rho \text{ such that } \text{Inf}(\rho) \cap G \neq \emptyset\}$$

Language accepted=Infinitely many b 's.

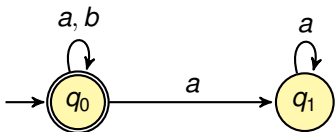
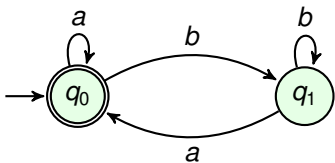
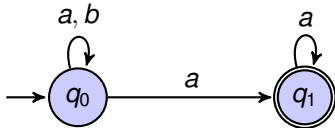
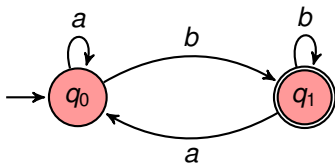
Comparing NFA and NBA



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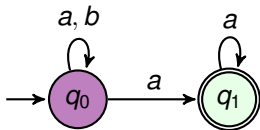
ω -Automata with Büchi Acceptance



- ▶ Left (T-B): Inf many b 's, Inf many a 's
- ▶ Right (T-B): Finitely many b 's, $(a + b)^\omega$

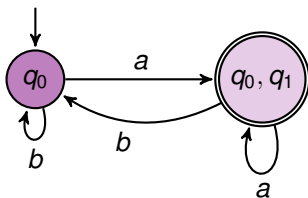
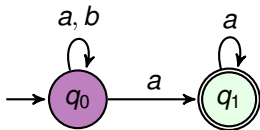
NBA and DBA

- ▶ Is every DBA as expressible as a NBA, like in the case of DFA and NFA?
- ▶ Can we do subset construction on NBA and obtain DBA?



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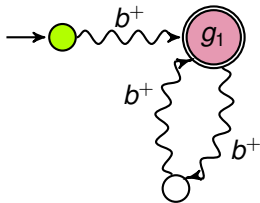


NBA and DBA

There does not exist a deterministic Büchi automata capturing the language finitely many a 's.

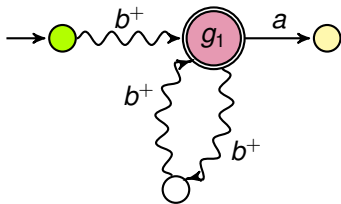
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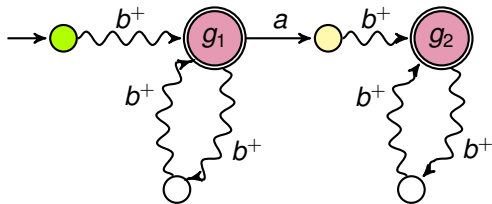
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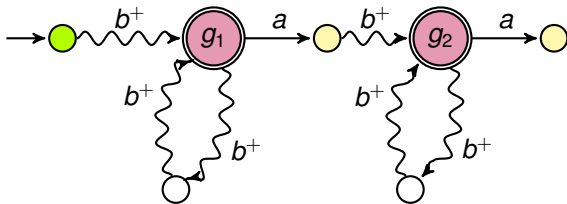
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